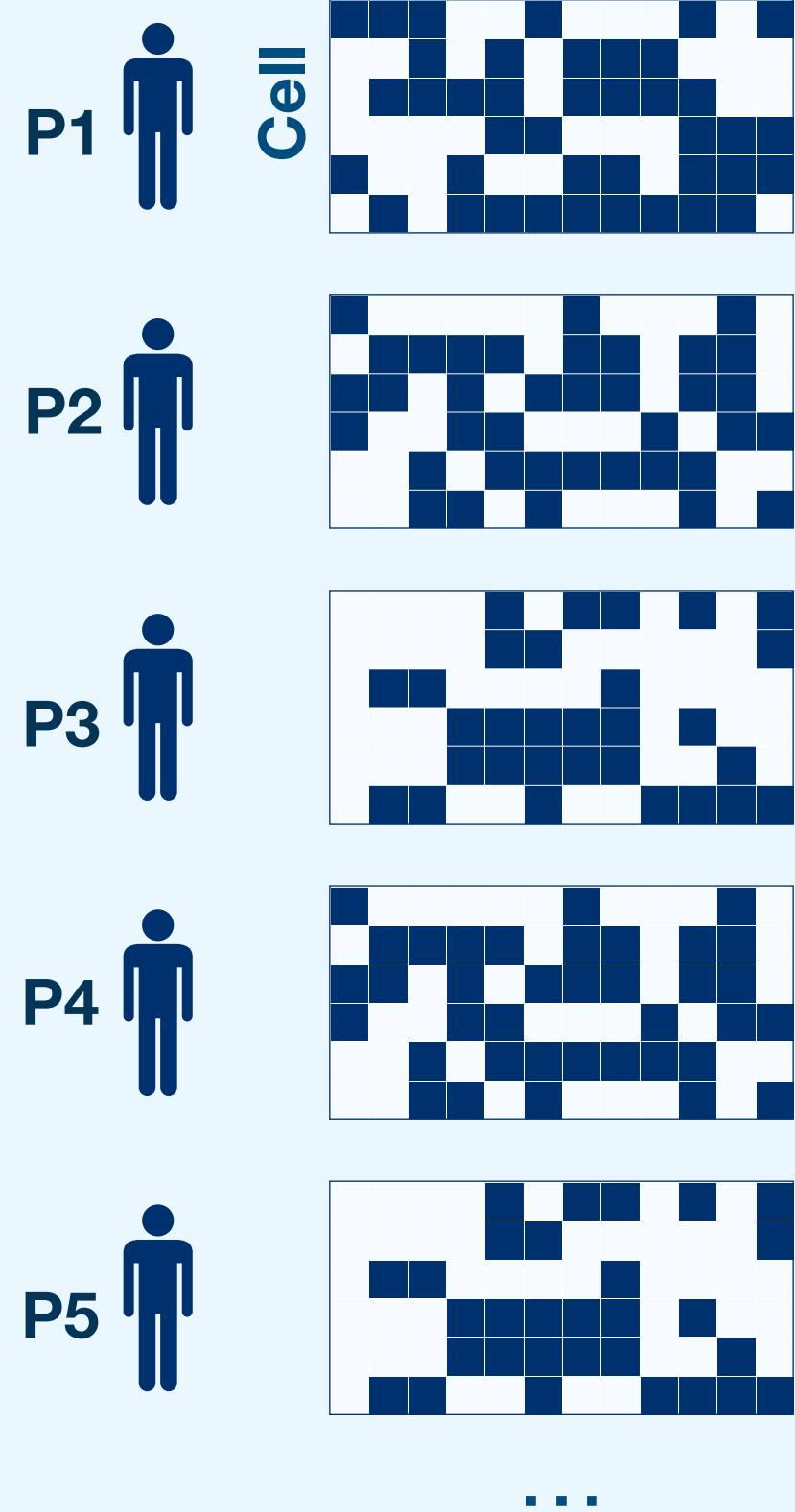


FALAFL (FAir muLti-sAmple Feature seLection)

Molecular Feature



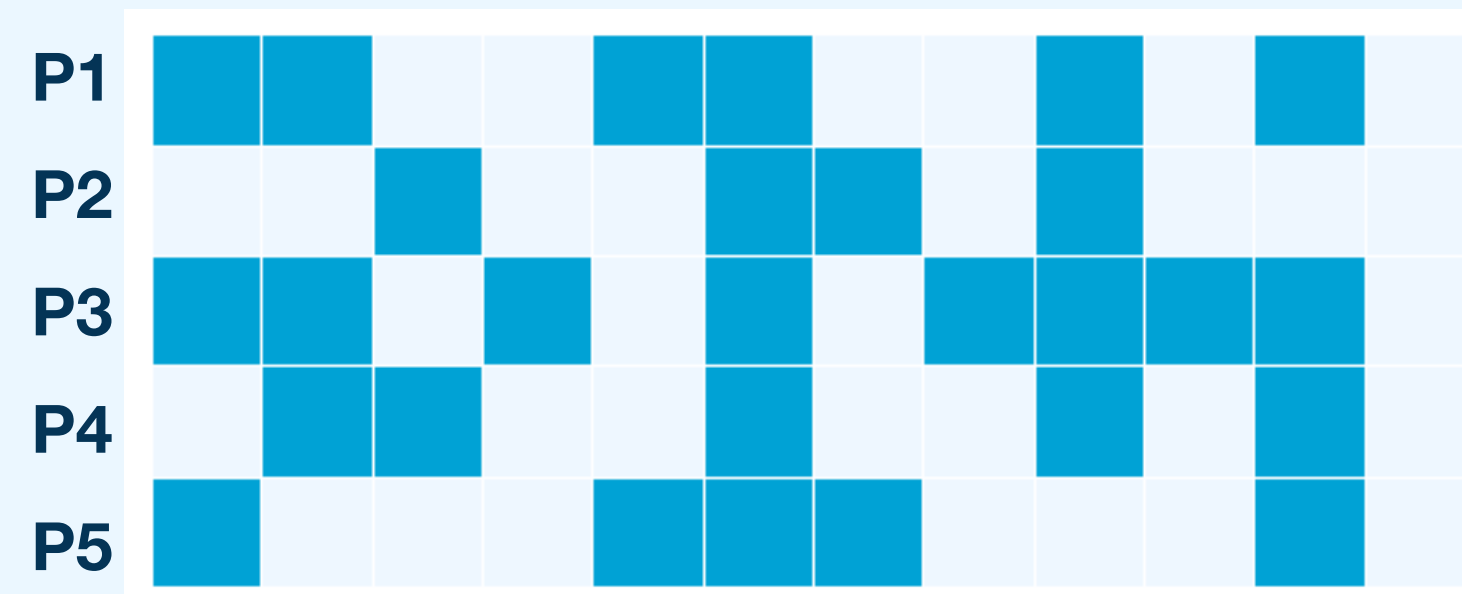
Sparse (binarized) cell-by-feature single-cell sequencing depth matrices for individual patients

Molecular Feature (e.g., CpG site)



S : Coverage per molecular feature per patient

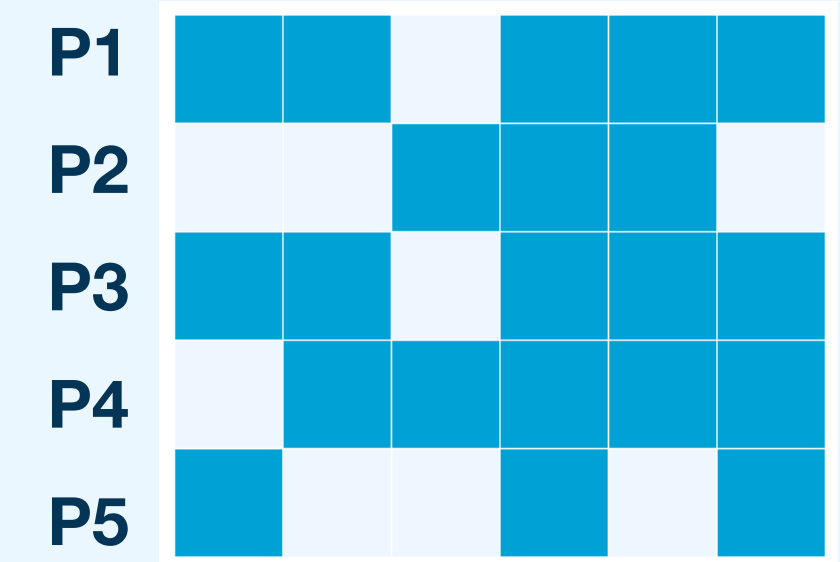
Molecular Feature (e.g., CpG site)



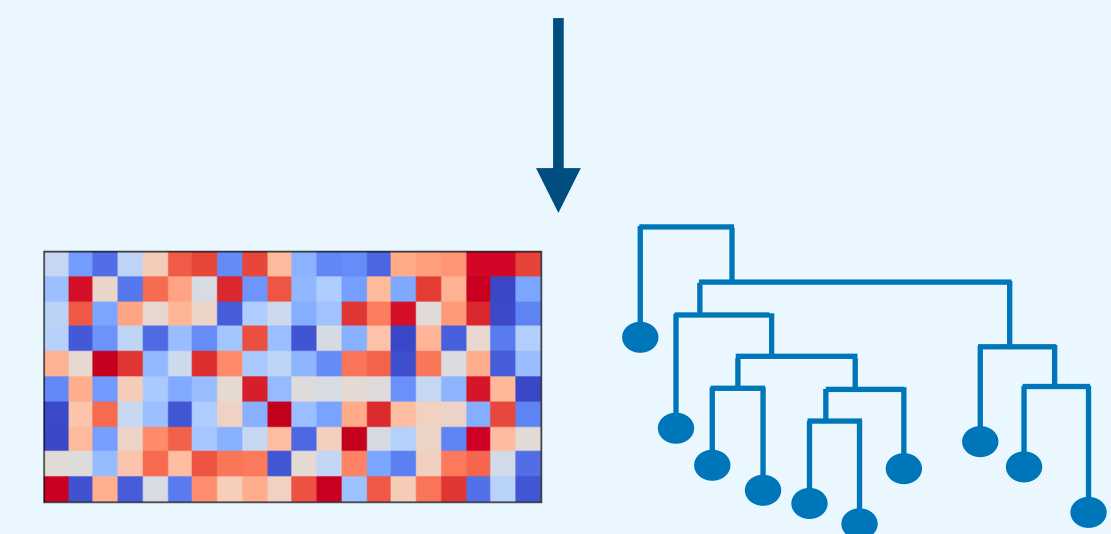
$\bar{S}$: Binarized coverage per molecular feature per patient sample based on user-defined coverage threshold p . Remove sites covered by less than user-defined k patients.

ILP-based optimization

Molecular Feature



FALAFL-selected optimal set of molecular features, with each patient harboring at least a user-defined fraction q of all selected molecular features.



Downstream analysis
e.g., evaluating the universality of CpG methylation dynamics